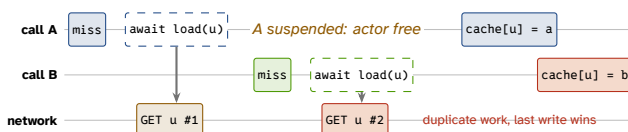
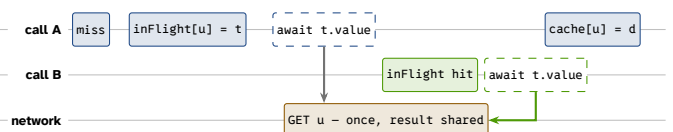


In one line: An **actor** is a reference type whose mutable state is guarded by a **serial executor**: one task at a time runs its isolated code, outsiders await. It is **reentrant** — at every *await* *inside* it other calls may run, so a check made before a suspension can be **stale** after. @MainActor is *one* global actor on the main thread: everything there is serialised with the UI.

Bug: two calls for the same URL interleave at the await



Fix: claim the work *before* the first await



Between two awaits an actor method is atomic. Across one, assume **anything** in the actor changed. Swift chose reentrancy to rule out deadlocks (an actor awaiting a call back into itself); the price is interleaving.

How it works

- Actor:** reference type, no inheritance, always Sendable. Inside: synchronous access to self. Outside: await even for a sync method (a possible hop). Arguments and results crossing the boundary must be Sendable (or sending).
- Executor:** a default actor is a serial executor whose jobs run on the **cooperative pool** — not a thread of its own. @MainActor's executor is the main thread. **Blocking** inside an actor (semaphore, sync I/O) starves the pool.
- Global actor:** @globalActor actor DB { static let shared = DB() }; @DB on types/functions puts *many* declarations in *one* domain. MainActor is one.
- Custom executor** (SE-0392, 5.9): nonisolated var unownedExecutor: UnownedSerialExecutor. Back an actor with a DispatchSerialQueue (iOS 17) to share a queue with a delegate API (AVCaptureSession); in its callbacks use assumeIsolated {} (traps if wrong) — preconditionIsolated() / assertIsolated() to check.
- nonisolated member:** outside the domain, sees only let Sendable state; needed for sync protocol requirements (Hashable, description).
- isolated parameter:** func f(_ b: isolated Bank) runs on b's executor, touching its state synchronously. **#isolation** (SE-0420): isolation: isolated (any Actor)? = #isolation = “run on my caller's actor” — a generic async helper that never hops, so non-Sendable closures are fine.
- @unchecked Sendable is acceptable** only when you synchronise every access (lock, Mutex, serial queue) or wrap a type documented thread-safe — state the invariant in a comment, run TSan.
- Mutex<State>** (Synchronization, iOS 18 *(unverified)*): ~Copyable, Sendable; withLock { \$0 ... } is synchronous, so the lock can never be held across an await. A final class with let m = Mutex(...) is plain Sendable — no @unchecked.

Why not always @MainActor?

	@MainActor	own actor	Mutex
runs on	main thread	coop. pool, serial	caller's thread
access	sync on main, else await	await	sync, blocks briefly
across await	reentrant	reentrant	impossible
parallel?	no — shares UI budget	vs other actors: yes	—
use for	UI state, view models	cache + network, DB	tiny sections, sync APIs

Other reentrancy fixes

- Mutate *before* suspending (reserve, then await).
- Re-check** state after every await; never carry a local copy of actor state across one.
- Move the invariant into a **synchronous** actor method.

Example — dedupe via the in-flight Task

```
actor ImageLoader {
    private var cache: [URL: Data] = [:]
    private var inFlight: [URL: Task<Data, any Error>] = [:]

    func data(for u: URL) async throws -> Data {
        if let hit = cache[u] { return hit }
        if let t = inFlight[u] { return try await t.value } // join
        let t = Task { try await URLSession.shared.data(from: u), 0 }
        inFlight[u] = t // claimed BEFORE any await
        defer { inFlight[u] = nil }
        let d = try await t.value // others may run here
        cache[u] = d // still correct after re-entry
        return d
    }
}

func pay(_ n: Int, from b: isolated Bank) { b.balance -= n }

import Synchronization
final class Metrics: Sendable { // no @unchecked needed
    private let hits = Mutex<String: Int>{[:]}
    func record(_ k: String) {
        hits.withLock { $0[k, default: 0] += 1 } }
}

actor Camera { // shares the capture queue
    let queue = DispatchSerialQueue(label: "camera")
    nonisolated var unownedExecutor: UnownedSerialExecutor {
        queue.asUnownedSerialExecutor() }
}
```

Traps

- ! “Actors make it thread-safe” — they remove *data* races, not *logic* races — reentrancy reorders state between awaits.
- ! Sync callers (delegate callback, description) cannot await an actor: Mutex, or assumeIsolated on the right executor.
- ! Waiting on actor work with a semaphore from the pool can deadlock: never block in async code.
- ! Everything on @MainActor = no parallelism and hangs; one actor per *resource*, not one per screen.

Remember

“An actor is a lock that lets go at every await.”